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ECA5801-IOE AND AI DRIVEN SYSTEM

EXPERIMENT 5:

a) Simulation and Performance Evaluation of RFID/NFC-Based Device Identification and IoE Data Integration using Node-RED, MQTT.

PROGRAM CODE:

```
# ================================================================
# EXPERIMENT (a)
# Simulation and Performance Evaluation of RFID/NFC-Based
# Device Identification and IoE Data Integration
# using Node-RED and MQTT
# ================================================================

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from datetime import datetime

np.random.seed(42)

# ================================================================
# 1. RFID/NFC DEVICE DATABASE
# ================================================================

devices = {
    "RFID001": {
        "Device": "Temperature Sensor",
```

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REG NO: 192512024

```
        "Location": "Laboratory",
        "Technology": "RFID"
    },

    "RFID002": {
        "Device": "Humidity Sensor",
        "Location": "Laboratory",
        "Technology": "RFID"
    },

    "NFC001": {
        "Device": "Smart Door Lock",
        "Location": "Main Entrance",
        "Technology": "NFC"
    },

    "NFC002": {
        "Device": "Control Unit",
        "Location": "Control Room",
        "Technology": "NFC"
    }
}



# ============================================================
# 2. RFID/NFC TAG SCANNING
# ============================================================
```

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REG NO: 192512024

```
print("=" * 70)
print("RFID/NFC DEVICE IDENTIFICATION")
print("=" * 70)

# Simulated RFID/NFC scans
scanned_tags = [
    "RFID001",
    "RFID002",
    "NFC001",
    "NFC002",
    "RFID001",
    "NFC001",
    "UNKNOWN001"
]

results = []

for tag in scanned_tags:

    timestamp = datetime.now().strftime("%H:%M:%S")

    if tag in devices:

        device = devices[tag]

        results.append([
            timestamp,
            tag,
```

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REG NO: 192512024

```
            device["Device"],
            device["Location"],
            device["Technology"],
            "SUCCESS"
        ])

        print(
            f"Tag: {tag} | "
            f"Device: {device['Device']} | "
            f"Status: SUCCESS"
        )

    else:

        results.append([
            timestamp,
            tag,
            "Unknown",
            "Unknown",
            "Unknown",
            "FAILED"
        ])

        print(
            f"Tag: {tag} | "
            f"Device: Unknown | "
            f"Status: FAILED"
        )
```

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REG NO: 192512024

```
# ============================================================
# 3. IDENTIFICATION TABLE
# ============================================================

identification_df = pd.DataFrame(
    results,
    columns=[
        "Time",
        "Tag ID",
        "Device",
        "Location",
        "Technology",
        "Status"
    ]
)

print("\nIDENTIFICATION RESULTS")
display(identification_df)


# ============================================================
# 4. GENERATE IoE DATA
# ============================================================

print("\n" + "=" * 70)
print("IoE DATA GENERATION")
```

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REG NO: 192512024

```
print("=" * 70)

ioe_data = []

for tag in scanned_tags:

    if tag not in devices:
        continue

    device = devices[tag]

    if device["Device"] == "Temperature Sensor":

        value = round(
            np.random.uniform(24, 35),
            2
        )

        unit = "°C"

    elif device["Device"] == "Humidity Sensor":

        value = round(
            np.random.uniform(40, 80),
            2
        )

        unit = "%"
```

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REG NO: 192512024

```
    elif device["Device"] == "Smart Door Lock":

        value = np.random.choice(
            ["OPEN", "CLOSED"]
        )

        unit = "State"

    else:

        value = np.random.randint(
            1,
            100
        )

        unit = "units"

    ioe_data.append([
        tag,
        device["Device"],
        device["Location"],
        value,
        unit
    ])


ioe_df = pd.DataFrame(
```

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REG NO: 192512024

```
    ioe_data,
    columns=[
        "Tag ID",
        "Device",
        "Location",
        "Value",
        "Unit"
    ]
)

display(ioe_df)



# ======================================================================
# 5. MQTT MESSAGE GENERATION
# ======================================================================

print("\n" + "=" * 70)
print("MQTT MESSAGE GENERATION")
print("=" * 70)

mqtt_messages = []

for _, row in ioe_df.iterrows():

    topic = (
        "IoE/"
        + row["Location"].replace(" ", "_")
```

NAME: SAKTHI DARSHINI V.

REG NO: 192512024

```
        + "/"
        + row["Device"].replace(" ", "_")
    )

    payload = (
        "{"
        f"'device_id':'{row['Tag ID']}', "
        f"'device':'{row['Device']}', "
        f"'value':'{row['Value']}', "
        f"'unit':'{row['Unit']}'"
        "}"
    )

    mqtt_messages.append([
        topic,
        payload
    ])

    print("MQTT PUBLISH")
    print("Topic   :", topic)
    print("Payload :", payload)
    print("-" * 60)


mqtt_df = pd.DataFrame(
    mqtt_messages,
    columns=[
        "MQTT Topic",
```

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REG NO: 192512024

```
        "MQTT Payload"
    ]
)



# ======================================================================
# 6. NODE-RED DATA PROCESSING
# ======================================================================

print("\n" + "=" * 70)
print("NODE-RED DATA PROCESSING")
print("=" * 70)

processed_data = []

for _, row in ioe_df.iterrows():

    device = row["Device"]
    value = row["Value"]

    if device == "Temperature Sensor":

        if value > 32:
            status = "HIGH"
        else:
            status = "NORMAL"

    elif device == "Humidity Sensor":
```

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REG NO: 192512024

```
        if value > 70:
            status = "HIGH"
        else:
            status = "NORMAL"

    elif device == "Smart Door Lock":

        status = value

    else:

        status = "ACTIVE"

    processed_data.append([
        row["Tag ID"],
        device,
        row["Location"],
        value,
        row["Unit"],
        status
    ])


processed_df = pd.DataFrame(
    processed_data,
    columns=[
        "Tag ID",
```

NAME: SAKTHI DARSHINI V.

REG NO: 192512024

```
        "Device",
        "Location",
        "Value",
        "Unit",
        "Node-RED Status"
    ]
)

display(processed_df)



# ================================================================
# 7. IDENTIFICATION SUCCESS RATE
# ================================================================

total_scans = len(scanned_tags)

successful_scans = sum(
    identification_df["Status"] == "SUCCESS"
)

failed_scans = total_scans - successful_scans

success_rate = (
    successful_scans /
    total_scans
) * 100
```

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REG NO: 192512024

```
print("\n" + "=" * 70)
print("PERFORMANCE RESULT")
print("=" * 70)

print("Total Scans       :", total_scans)
print("Successful Scans  :", successful_scans)
print("Failed Scans      :", failed_scans)
print(
    "Identification Success Rate:",
    round(success_rate, 2),
    "%"
)



# ================================================================
# 8. IDENTIFICATION PERFORMANCE GRAPH
# ================================================================

plt.figure(figsize=(8, 5))

plt.bar(
    ["Successful", "Failed"],
    [successful_scans, failed_scans]
)

plt.xlabel("Identification Result")
plt.ylabel("Number of Tags")
plt.title("RFID/NFC Device Identification Performance")
```

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REG NO: 192512024

```
plt.grid(
    axis="y",
    linestyle="--",
    alpha=0.5
)

plt.show()



# ======================================================================
# 9. FINAL DATA INTEGRATION RESULT
# ======================================================================

print("\n" + "=" * 70)
print("FINAL IoE DATA INTEGRATION")
print("=" * 70)

print("""
RFID/NFC TAG
    ↓
DEVICE IDENTIFICATION
    ↓
IoE DATA GENERATION
    ↓
MQTT TOPIC + PAYLOAD
    ↓
NODE-RED PROCESSING
```

NAME: SAKTHI DARSHINI V.

REG NO: 192512024

```
    ↓
PROCESSED IoE DATA
""")


print(
    "Total MQTT Messages:",
    len(mqtt_messages)
)


print(
    "Identification Success Rate:",
    round(success_rate, 2),
    "%"
)
```

OUTPUT:

```
======================================================================
RFID/NFC DEVICE IDENTIFICATION
======================================================================
Tag: RFID001 | Device: Temperature Sensor | Status: SUCCESS
Tag: RFID002 | Device: Humidity Sensor | Status: SUCCESS
Tag: NFC001 | Device: Smart Door Lock | Status: SUCCESS
Tag: NFC002 | Device: Control Unit | Status: SUCCESS
Tag: RFID001 | Device: Temperature Sensor | Status: SUCCESS
Tag: NFC001 | Device: Smart Door Lock | Status: SUCCESS
Tag: UNKNOWN001 | Device: Unknown | Status: FAILED

IDENTIFICATION RESULTS
```

	Time	Tag ID	Device	Location	Technology	Status
0	05:12:32	RFID001	Temperature Sensor	Laboratory	RFID	SUCCESS
1	05:12:32	RFID002	Humidity Sensor	Laboratory	RFID	SUCCESS
2	05:12:32	NFC001	Smart Door Lock	Main Entrance	NFC	SUCCESS
3	05:12:32	NFC002	Control Unit	Control Room	NFC	SUCCESS
4	05:12:32	RFID001	Temperature Sensor	Laboratory	RFID	SUCCESS
5	05:12:32	NFC001	Smart Door Lock	Main Entrance	NFC	SUCCESS
6	05:12:32	UNKNOWN001	Unknown	Unknown	Unknown	FAILED

NAME: SAKTHI DARSHINI V.

REG NO: 192512024

```
============================================================
IoE DATA GENERATION
============================================================
```

	Tag ID	Device	Location	Value	Unit
0	RFID001	Temperature Sensor	Laboratory	28.12	°C
1	RFID002	Humidity Sensor	Laboratory	78.03	%
2	NFC001	Smart Door Lock	Main Entrance	OPEN	State
3	NFC002	Control Unit	Control Room	72	units
4	RFID001	Temperature Sensor	Laboratory	30.59	°C
5	NFC001	Smart Door Lock	Main Entrance	OPEN	State

```
======================================================================
MQTT MESSAGE GENERATION
======================================================================
MQTT PUBLISH
Topic   : IoE/Laboratory/Temperature_Sensor
Payload : {'device_id':'RFID001', 'device':'Temperature Sensor', 'value':'28.12', 'unit':'°C'}
------------------------------------------------------------
MQTT PUBLISH
Topic   : IoE/Laboratory/Humidity_Sensor
Payload : {'device_id':'RFID002', 'device':'Humidity Sensor', 'value':'78.03', 'unit':'%'}
------------------------------------------------------------
MQTT PUBLISH
Topic   : IoE/Main_Entrance/Smart_Door_Lock
Payload : {'device_id':'NFC001', 'device':'Smart Door Lock', 'value':'OPEN', 'unit':'State'}
------------------------------------------------------------
MQTT PUBLISH
Topic   : IoE/Control_Room/Control_Unit
Payload : {'device_id':'NFC002', 'device':'Control Unit', 'value':'72', 'unit':'units'}
------------------------------------------------------------
MQTT PUBLISH
Topic   : IoE/Laboratory/Temperature_Sensor
Payload : {'device_id':'RFID001', 'device':'Temperature Sensor', 'value':'30.59', 'unit':'°C'}
------------------------------------------------------------
MQTT PUBLISH
Topic   : IoE/Main_Entrance/Smart_Door_Lock
Payload : {'device_id':'NFC001', 'device':'Smart Door Lock', 'value':'OPEN', 'unit':'State'}
------------------------------------------------------------
```

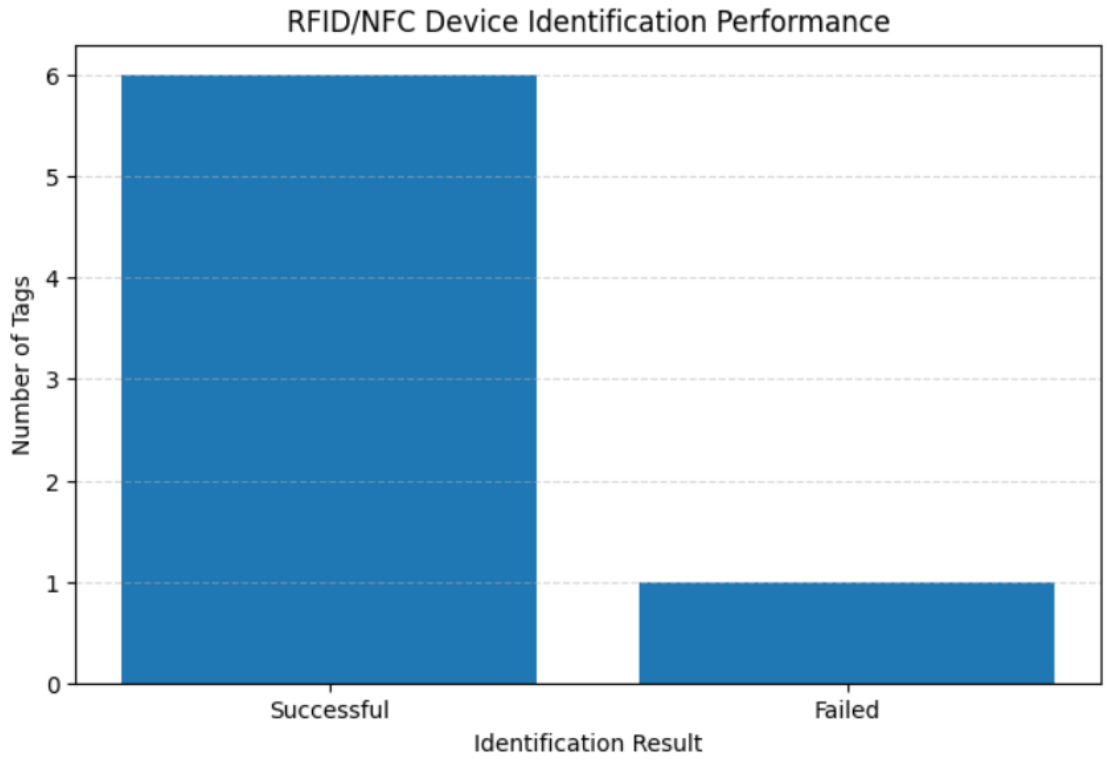
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REG NO: 192512024

```
======================================================================
NODE-RED DATA PROCESSING
======================================================================
```

	Tag ID	Device	Location	Value	Unit	Node-RED Status
0	RFID001	Temperature Sensor	Laboratory	28.12	°C	NORMAL
1	RFID002	Humidity Sensor	Laboratory	78.03	%	HIGH
2	NFC001	Smart Door Lock	Main Entrance	OPEN	State	OPEN
3	NFC002	Control Unit	Control Room	72	units	ACTIVE
4	RFID001	Temperature Sensor	Laboratory	30.59	°C	NORMAL
5	NFC001	Smart Door Lock	Main Entrance	OPEN	State	OPEN

```
======================================================================
PERFORMANCE RESULT
======================================================================
Total Scans       : 7
Successful Scans  : 6
Failed Scans      : 1
Identification Success Rate: 85.71 %
```



NAME: SAKTHI DARSHINI V.

REG NO: 192512024

```
================================================================================
FINAL IoE DATA INTEGRATION
================================================================================

RFID/NFC TAG
     ↓
DEVICE IDENTIFICATION
     ↓
IoE DATA GENERATION
     ↓
MQTT TOPIC + PAYLOAD
     ↓
NODE-RED PROCESSING
     ↓
PROCESSED IoE DATA

Total MQTT Messages: 6
Identification Success Rate: 85.71 %
```

NAME: SAKTHI DARSHINI V.

REG NO: 192512024

EXPERIMENT 5:

b) Modelling and Simulation of a 4-Layer IoE Architecture (Perception–Network–Middleware–Application) using Node-RED, MQTT and a Cloud Dashboard

PROGRAM:

```
# ================================================================
# Modelling and Simulation of a 4-Layer IoE Architecture
# (Perception - Network - Middleware - Application)
# using Node-RED, MQTT and Cloud Dashboard
# ================================================================

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

np.random.seed(42)

# ================================================================
# 1. IoE DEVICE CONFIGURATION
# ================================================================

devices = [
    "Temperature Sensor",
    "Humidity Sensor",
    "Light Sensor",
    "Motion Sensor"
```

NAME: SAKTHI DARSHINI V.

REG NO: 192512024

```
]

locations = [
    "Laboratory",
    "Laboratory",
    "Classroom",
    "Classroom"
]



# ======================================================================
# 2. LAYER 1 — PERCEPTION LAYER
# ======================================================================

print("=" * 75)
print("LAYER 1 — PERCEPTION LAYER")
print("=" * 75)

perception_data = []

for i in range(len(devices)):

    device = devices[i]
    location = locations[i]

    if device == "Temperature Sensor":

        value = round(
```

NAME: SAKTHI DARSHINI V.

REG NO: 192512024

```
        np.random.uniform(24, 35),
        2
    )

    unit = "°C"

elif device == "Humidity Sensor":

    value = round(
        np.random.uniform(40, 80),
        2
    )

    unit = "%"

elif device == "Light Sensor":

    value = round(
        np.random.uniform(100, 900),
        2
    )

    unit = "lux"

else:

    value = np.random.choice(
        [0, 1]
```

NAME: SAKTHI DARSHINI V.

REG NO: 192512024

```
        )

        unit = "Motion"


    perception_data.append([
        device,
        location,
        value,
        unit
    ])


perception_df = pd.DataFrame(
    perception_data,
    columns=[
        "Device",
        "Location",
        "Sensor Value",
        "Unit"
    ]
)

print("Sensors collect physical-world data:")

display(perception_df)
```

NAME: SAKTHI DARSHINI V.

REG NO: 192512024

```
# ============================================================
# 3. LAYER 2 — NETWORK LAYER
# ============================================================

print("\n" + "=" * 75)
print("LAYER 2 — NETWORK LAYER")
print("=" * 75)

mqtt_data = []

for _, row in perception_df.iterrows():

    topic = (
        "IoE/"
        + row["Location"].replace(" ", "_")
        + "/"
        + row["Device"].replace(" ", "_")
    )

    payload = (
        "{"
        f"'device':'{row['Device']}', "
        f"'location':'{row['Location']}', "
        f"'value':{row['Sensor Value']}, "
        f"'unit':'{row['Unit']}'"
        "}"
    )
```

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REG NO: 192512024

```
    mqtt_data.append([
        topic,
        payload
    ])

    print("MQTT PUBLISH")
    print("Topic   :", topic)
    print("Payload :", payload)
    print("-" * 60)


mqtt_df = pd.DataFrame(
    mqtt_data,
    columns=[
        "MQTT Topic",
        "MQTT Payload"
    ]
)

display(mqtt_df)


# ============================================================
# 4. LAYER 3 — MIDDLEWARE LAYER
# ============================================================

print("\n" + "=" * 75)
print("LAYER 3 — MIDDLEWARE LAYER")
```

NAME: SAKTHI DARSHINI V.

REG NO: 192512024

```
print("=" * 75)

middleware_data = []

for _, row in perception_df.iterrows():

    device = row["Device"]
    value = row["Sensor Value"]

    # Node-RED processing rules

    if device == "Temperature Sensor":

        if value > 32:
            status = "ALERT"
        else:
            status = "NORMAL"

    elif device == "Humidity Sensor":

        if value > 70:
            status = "ALERT"
        else:
            status = "NORMAL"

    elif device == "Light Sensor":

        if value < 200:
```

NAME: SAKTHI DARSHINI V.

REG NO: 192512024

```
            status = "LOW LIGHT"
        else:
            status = "NORMAL"

    elif device == "Motion Sensor":

        if value == 1:
            status = "MOTION DETECTED"
        else:
            status = "NO MOTION"

    middleware_data.append([
        device,
        row["Location"],
        value,
        row["Unit"],
        status
    ])


middleware_df = pd.DataFrame(
    middleware_data,
    columns=[
        "Device",
        "Location",
        "Value",
        "Unit",
        "Node-RED Status"
```

NAME: SAKTHI DARSHINI V.

REG NO: 192512024

```
    ]
)

display(middleware_df)



# ============================================================================
# 5. LAYER 4 — APPLICATION LAYER
# ============================================================================

print("\n" + "=" * 75)
print("LAYER 4 — APPLICATION / CLOUD DASHBOARD")
print("=" * 75)

dashboard_df = middleware_df.copy()

display(dashboard_df)



# ============================================================================
# 6. DASHBOARD GRAPH
# ============================================================================

plt.figure(figsize=(10, 5))

plt.bar(
    dashboard_df["Device"],
    dashboard_df["Value"]
```

NAME: SAKTHI DARSHINI V.

REG NO: 192512024

```
)

plt.xlabel("IoE Device")
plt.ylabel("Sensor Value")

plt.title(
    "IoE Cloud Dashboard - Device Data"
)

plt.xticks(
    rotation=20
)

plt.grid(
    axis="y",
    linestyle="--",
    alpha=0.5
)

plt.show()



# ============================================================
# 7. IoE FOUR-LAYER DATA FLOW
# ============================================================

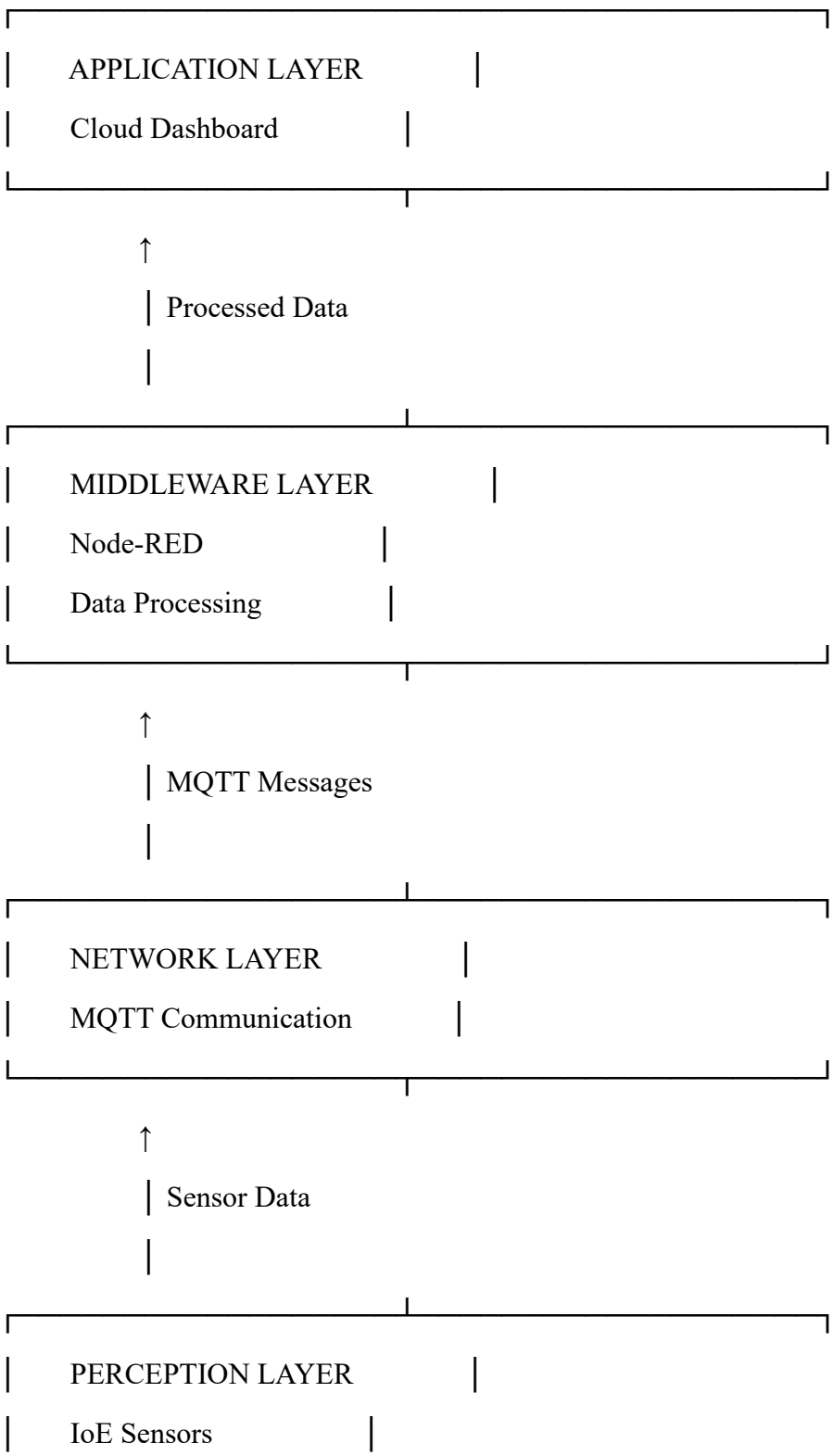
print("\n" + "=" * 75)
print("4-LAYER IoE DATA FLOW")
```

NAME: SAKTHI DARSHINI V.

REG NO: 192512024

```
print("=" * 75)

print("""
```



NAME: SAKTHI DARSHINI V.

REG NO: 192512024

```
└──────────────────────────────────────────────────────────────┘
""")


# ============================================================
# 8. DATA FLOW PERFORMANCE
# ============================================================

total_devices = len(perception_df)

mqtt_messages = len(mqtt_df)

processed_messages = len(middleware_df)

dashboard_records = len(dashboard_df)



print("=" * 75)
print("DATA FLOW RESULTS")
print("=" * 75)

print(
    "Perception Layer Records :", total_devices
)

print(
    "MQTT Messages Transmitted:",
    mqtt_messages
```

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REG NO: 192512024

```
)

print(
    "Node-RED Messages Processed:",
    processed_messages
)

print(
    "Dashboard Records:",
    dashboard_records
)



# ================================================================
# 9. DATA FLOW SUCCESS RATE
# ================================================================

if total_devices > 0:

    network_success = (
        mqtt_messages /
        total_devices
    ) * 100

    middleware_success = (
        processed_messages /
        mqtt_messages
    ) * 100
```

NAME: SAKTHI DARSHINI V.

REG NO: 192512024

```
    dashboard_success = (
        dashboard_records /
        processed_messages
    ) * 100

else:

    network_success = 0
    middleware_success = 0
    dashboard_success = 0


print("\nData Flow Performance")

print(
    "Perception → Network:",
    round(network_success, 2),
    "%"
)

print(
    "Network → Middleware:",
    round(middleware_success, 2),
    "%"
)

print(
```

NAME: SAKTHI DARSHINI V.

REG NO: 192512024

```
    "Middleware → Application:",
    round(dashboard_success, 2),
    "%"
)



# ================================================================
# 10. DATA FLOW PERFORMANCE GRAPH
# ================================================================

layers = [
    "Perception",
    "Network",
    "Middleware",
    "Application"
]

records = [
    total_devices,
    mqtt_messages,
    processed_messages,
    dashboard_records
]

plt.figure(figsize=(9, 5))

plt.bar(
    layers,
```

NAME: SAKTHI DARSHINI V.

REG NO: 192512024

```
    records
)

plt.xlabel("IoE Layer")
plt.ylabel("Number of Data Records")

plt.title(
    "Data Flow Through 4-Layer IoE Architecture"
)

plt.grid(
    axis="y",
    linestyle="--",
    alpha=0.5
)

plt.show()



# ============================================================
# 11. FINAL RESULT
# ============================================================

print("\n" + "=" * 75)
print("FINAL EXPERIMENT RESULT")
print("=" * 75)

print("""
```

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REG NO: 192512024

```
PERCEPTION
   ↓
Sensors collect physical data
   ↓
NETWORK
   ↓
MQTT transfers IoE messages
   ↓
MIDDLEWARE
   ↓
Node-RED processes the messages
   ↓
APPLICATION
   ↓
Cloud Dashboard displays the information
""")

print(
   "Total IoE Devices:",
   total_devices
)

print(
   "Total MQTT Messages:",
   mqtt_messages
)

print(
```

NAME: SAKTHI DARSHINI V.

REG NO: 192512024

```
    "Total Processed Messages:",
    processed_messages
)

print(
    "Total Dashboard Records:",
    dashboard_records
)
```

OUTPUT :

```
========================================================================
LAYER 1 — PERCEPTION LAYER
========================================================================
Sensors collect physical-world data:
```

	Device	Location	Sensor Value	Unit
0	Temperature Sensor	Laboratory	28.12	°C
1	Humidity Sensor	Laboratory	78.03	%
2	Light Sensor	Classroom	685.60	lux
3	Motion Sensor	Classroom	0.00	Motion

```
========================================================================
LAYER 2 — NETWORK LAYER
========================================================================
MQTT PUBLISH
Topic   : IoE/Laboratory/Temperature_Sensor
Payload : {'device':'Temperature Sensor', 'location':'Laboratory', 'value':28.12, 'unit':'°C'}
------------------------------------------------------------
MQTT PUBLISH
Topic   : IoE/Laboratory/Humidity_Sensor
Payload : {'device':'Humidity Sensor', 'location':'Laboratory', 'value':78.03, 'unit':'%'}
------------------------------------------------------------
MQTT PUBLISH
Topic   : IoE/Classroom/Light_Sensor
Payload : {'device':'Light Sensor', 'location':'Classroom', 'value':685.6, 'unit':'lux'}
```

NAME: SAKTHI DARSHINI V.

REG NO: 192512024

	MQTT Topic	MQTT Payload
0	IoE/Laboratory/Temperature_Sensor	{'device':'Temperature Sensor', 'location':'La...
1	IoE/Laboratory/Humidity_Sensor	{'device':'Humidity Sensor', 'location':'Labor...
2	IoE/Classroom/Light_Sensor	{'device':'Light Sensor', 'location':'Classroo...
3	IoE/Classroom/Motion_Sensor	{'device':'Motion Sensor', 'location':'Classro...

```
================================================================================
LAYER 3 — MIDDLEWARE LAYER
================================================================================
```

	Device	Location	Value	Unit	Node-RED Status
0	Temperature Sensor	Laboratory	28.12	°C	NORMAL
1	Humidity Sensor	Laboratory	78.03	%	ALERT
2	Light Sensor	Classroom	685.60	lux	NORMAL
3	Motion Sensor	Classroom	0.00	Motion	NO MOTION

```
============================================================
LAYER 4 — APPLICATION / CLOUD DASHBOARD
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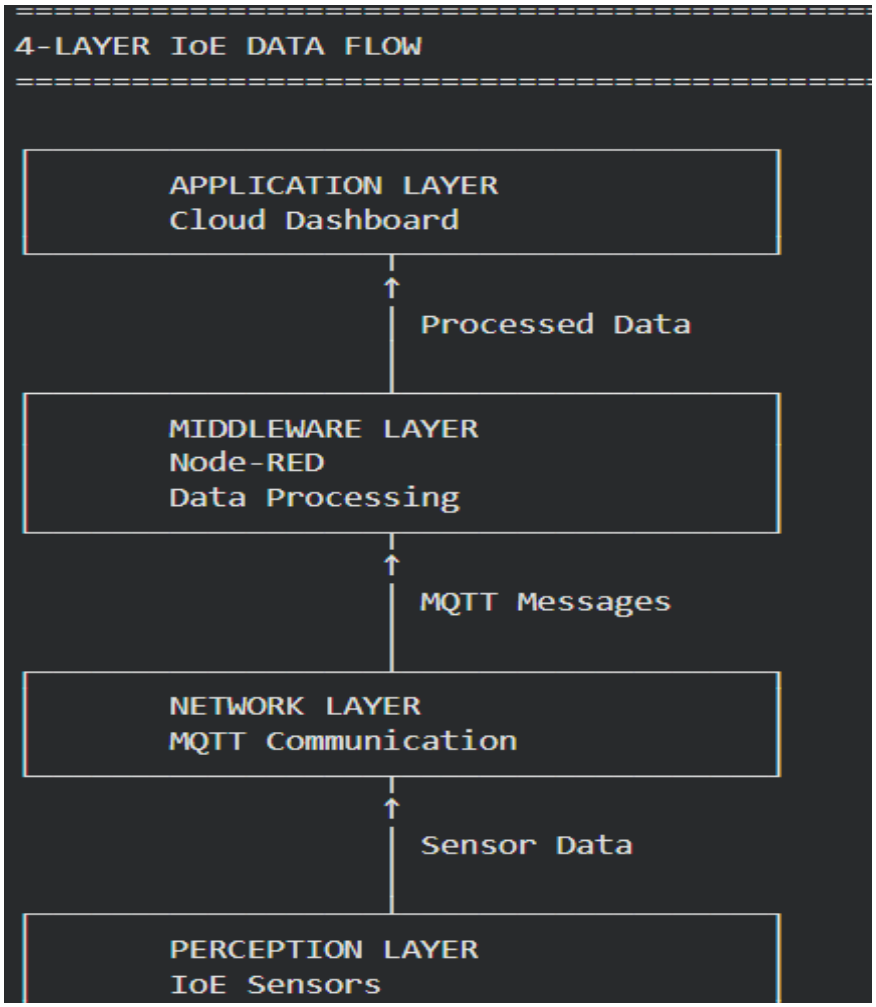
	Device	Location	Value	Unit	Node-RED Status
0	Temperature Sensor	Laboratory	28.12	°C	NORMAL
1	Humidity Sensor	Laboratory	78.03	%	ALERT
2	Light Sensor	Classroom	685.60	lux	NORMAL
3	Motion Sensor	Classroom	0.00	Motion	NO MOTION

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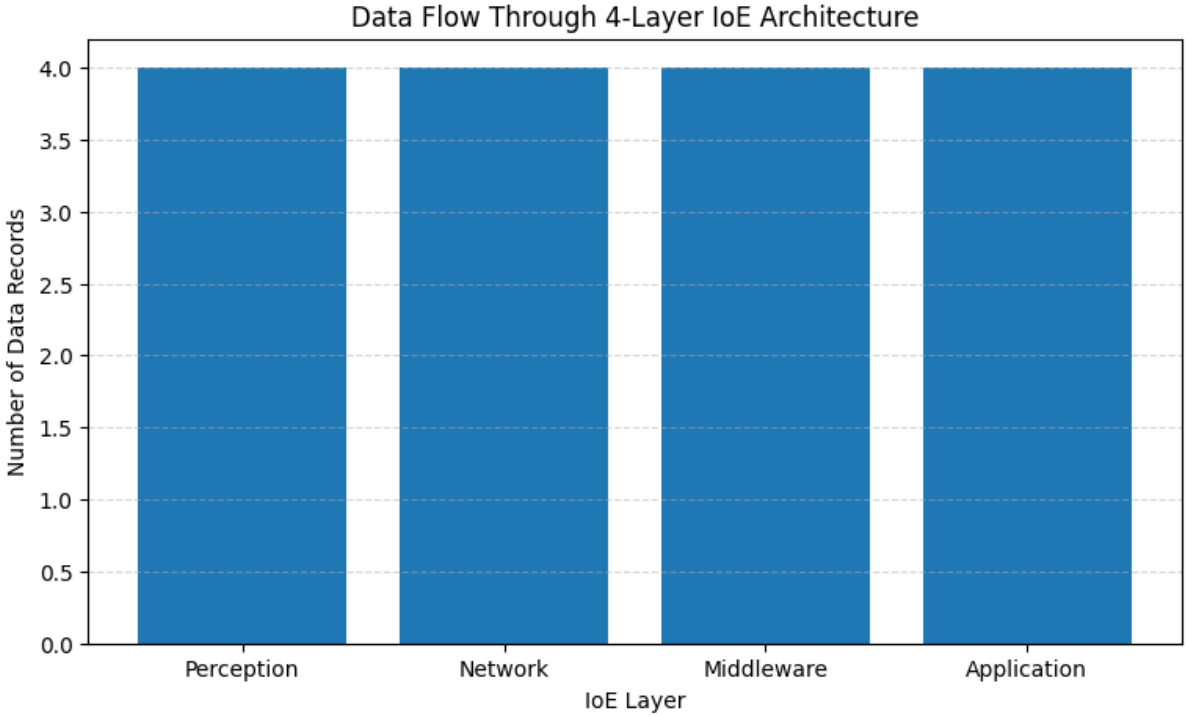
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DATA FLOW RESULTS
Perception Layer Records : 4
MQTT Messages Transmitted: 4
Node-RED Messages Processed: 4
Dashboard Records: 4

Data Flow Performance
Perception → Network: 100.0 %
Network → Middleware: 100.0 %
Middleware → Application: 100.0 %
```



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FINAL EXPERIMENT RESULT
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PERCEPTION
    ↓
Sensors collect physical data
    ↓
NETWORK
    ↓
MQTT transfers IoE messages
    ↓
MIDDLEWARE
    ↓
Node-RED processes the messages
    ↓
APPLICATION
    ↓
Cloud Dashboard displays the information

Total IoE Devices: 4
Total MQTT Messages: 4
Total Processed Messages: 4
Total Dashboard Records: 4
```

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